SalonEase: Salon Appointment and Management System

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1. Problem Statement

Many small salons still manage bookings in paper notebooks. Customers call or visit, and the receptionist writes a name and a time. As a result, two customers are sometimes booked with the same stylist at the same time, customer and payment records are scattered, the owner cannot easily see how much the salon earns each day or month, and nobody knows which services are most popular or how many appointments are cancelled. Missed and cancelled appointments also cost money: a 2025 UK survey found hair and beauty businesses lose about 7% of monthly revenue to them (Fresha, 2026), and a salon in a Malaysian case study had double-booked chairs because it used a paper diary (Fadil, 2025).

2. Proposed Solution

SalonEase is an Oracle database with PL/SQL that organises salon work in one place. The receptionist registers the customer, selects the service and stylist, chooses a date and time, and records payment after the service. The system prevents double booking, calculates the total cost of services, records completed and cancelled appointments, and reports daily and monthly revenue and the most popular services. It has three parts: a user interface for staff, the Oracle database, and a dashboard.

3. Data Source

The main dataset is the Hair Salon No-Show Dataset on Kaggle (author: frederickferguson), from a hair salon in Toronto, March to July 2018. The file used has 1,952 booking records and 22 columns (booking day, time, service category, stylist, details of the client's previous visit, and a no-show flag); 11.5% of bookings are no-shows. It has no client names, phone numbers or dates, and stylist first names will be replaced with codes. Licence: CC0: Public Domain. Because it comes from Canada and has no client ID, a simple survey of salons in Kigali will supply local services and prices, and generated sample records (clearly labelled) will fill CLIENT, PAYMENT and the real dates.

4. Planned Tables

CLIENT, STYLIST, SERVICE, STYLIST_SERVICE, STYLIST_SCHEDULE, APPOINTMENT, APPOINTMENT_SERVICE, PAYMENT and AUDIT_LOG, normalized to Third Normal Form (3NF). A client has many appointments; an appointment has one stylist, many services and its payments.

5. Planned PL/SQL Features

- **Procedures:** register_client, book_appointment, cancel_appointment, complete_appointment and record_payment, used by the receptionist for each step of the workflow.
- **Functions:** is_stylist_available checks that a stylist is free before saving a booking; calc_appointment_total adds up the price of all services in an appointment.
- **Cursors:** loop over completed appointments to produce daily and monthly revenue and a popular-services ranking.
- **Package:** salonease_pkg groups the booking, payment and report logic.

- **Triggers:** one blocks two appointments for the same stylist at overlapping times; one blocks payment for a cancelled appointment; a restriction rule blocks bookings in the past or outside opening hours; an audit trigger records every change and cancellation in AUDIT_LOG.
- **Exceptions:** custom errors such as stylist_not_available and invalid_appointment_time, shown as clear messages to staff.

6. Innovation

Option: User interface combined with Analytics dashboard. **Tool:** Oracle APEX, connected to the live Oracle database. Salon staff register customers, book appointments, manage services and record payments in the app, and each action calls a PL/SQL procedure. A dashboard page shows daily bookings, monthly revenue, appointment statistics (completed, cancelled) and popular services. Business rules stay in PL/SQL; APEX only displays and sends data. It will be demonstrated live in Phase VII and documented in the innovation/ folder.

7. References (APA)

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